

Support and Maintenance

ctrlaltcrew — Post-Launch Support Reference

This document explains what happens after a project launches: the initial warranty period, ongoing support and maintenance options, and how AI/LLM systems in particular are kept up to date over time.

1. Post-Launch Warranty Period

Every project includes an initial warranty period after deployment, during which defects related to the originally agreed scope are fixed at no additional cost. This covers bugs and issues that trace back to the delivered work, not new feature requests or changes to requirements made after launch.

2. Bug Fixing

Bug fixing refers to correcting unintended behavior in the delivered application — for example, a broken form, a display issue, or an incorrect calculation. Bugs identified during the warranty period are addressed as part of that coverage. After the warranty period, bug fixing is covered under an ongoing support plan or handled as a standalone request.

3. Maintenance

Maintenance covers the routine work needed to keep an application running reliably over time: applying security patches, updating dependencies/libraries, monitoring server or hosting health, and ensuring compatibility as browsers, operating systems, or third-party APIs change.

4. Updates and Feature Enhancements

Updates and enhancements refer to new functionality added after the initial launch — for example, adding a new page, a new integration, or a new feature to an existing AI assistant. These are scoped and estimated similarly to new project work, sized to the specific enhancement requested, and can be handled individually or bundled into a support plan.

5. Monitoring

For applications on an active support plan, basic uptime and error monitoring can be put in place so issues are identified proactively rather than only when a user reports them. For AI/LLM applications, this can include monitoring for unusual response patterns or repeated failed retrievals.

6. AI/LLM Maintenance

AI-powered applications require some maintenance considerations that traditional software does not:

- Periodic review of chatbot or assistant conversations (where appropriate and privacy-compliant) to identify common failure patterns or unanswered questions.
- Prompt and configuration adjustments as the underlying LLM provider updates its models.
- Monitoring for 'drift' — cases where the assistant's answers become less accurate as the client's business, products, or documents change without a corresponding knowledge-base update.

7. Knowledge-Base Updates for RAG Systems

RAG-based assistants and knowledge-base assistants answer questions using a defined set of source documents. When those documents change — new policies, updated product information, revised FAQs — the underlying knowledge base needs to be refreshed so the assistant's answers stay accurate. This involves re-ingesting updated or new documents, regenerating embeddings for changed content, and removing outdated content from the retrieval index.

Clients with frequently changing content (e.g., weekly policy updates) typically need more frequent knowledge-base refreshes than clients with largely static documentation. Update frequency and responsibility (client-managed vs. ctrlaltcrew-managed updates) are agreed as part of the support plan.

8. What Can Cause Additional Maintenance Costs

- Frequent content or document changes requiring repeated knowledge-base updates for AI/RAG systems.
- Third-party API or platform changes that break existing integrations and require rework.
- Requests for new features or enhancements beyond the original project scope.
- Significant traffic or usage growth that requires infrastructure scaling beyond the original setup.
- Migrating the application to a different hosting provider or technology stack after launch.

9. Support Plan Structure

After the initial warranty period, clients can choose to continue with an ongoing support arrangement rather than handling every future request as a one-off project. Typical support arrangements are scoped around three factors: response-time expectations, the volume of expected update/fix requests per month, and whether knowledge-base updates (for AI systems) are included. Exact support plan pricing is agreed per client based on these factors and is not fixed in this document; clients should discuss specific support needs during the post-launch stage of their project.

10. Summary

Support Area	Included in Warranty Period	Typically Requires Ongoing Plan
Bug fixes tied to original scope	Yes	No (unless after warranty ends)
Security/dependency updates	Limited	Yes

Support Area	Included in Warranty Period	Typically Requires Ongoing Plan
New features/enhancements	No	Yes (scoped individually or via plan)
Uptime/error monitoring	No	Yes
RAG knowledge-base refresh	No	Yes, if content changes over time